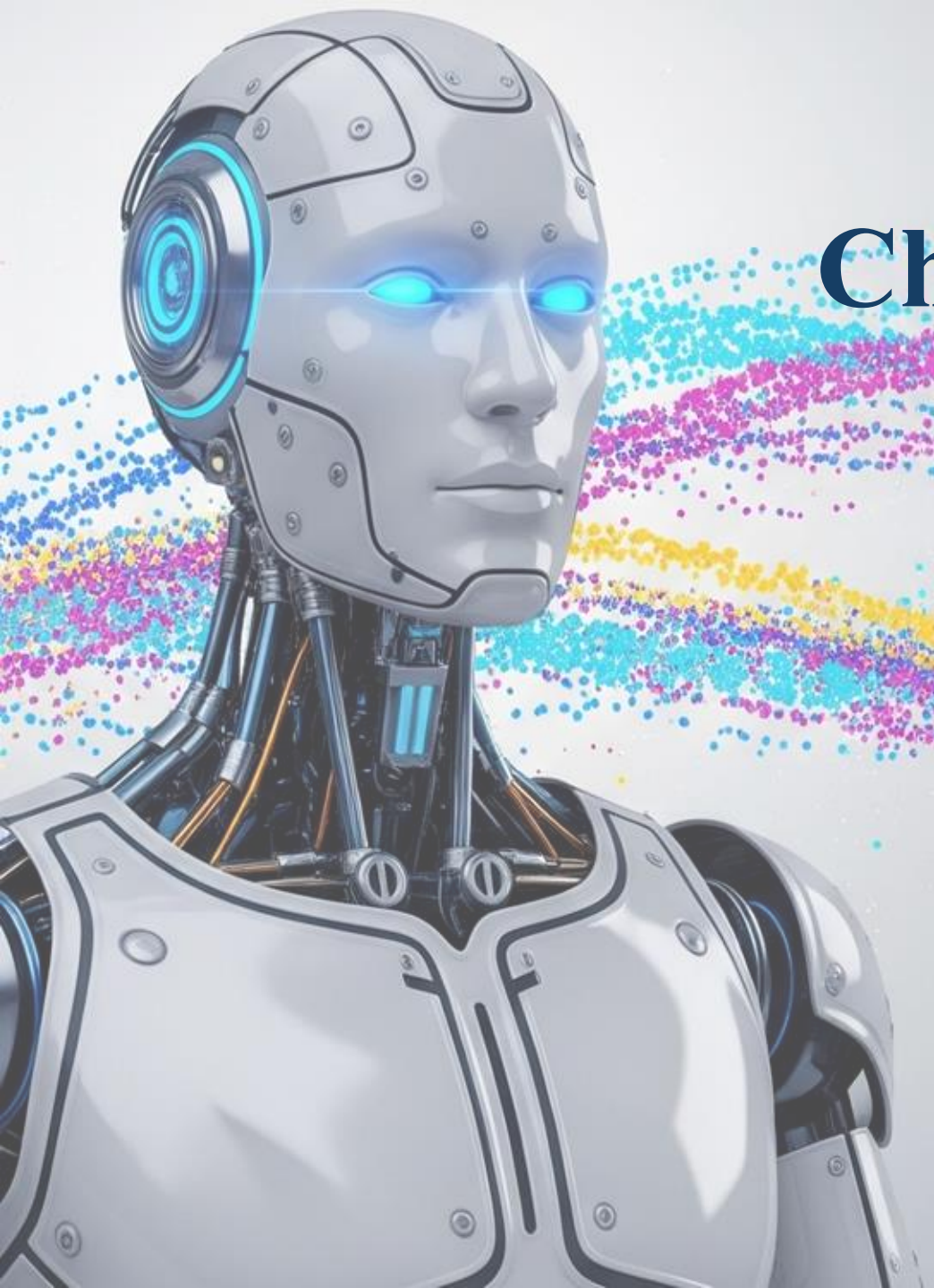


Chapter 4: Semi-Supervised Learning

Ph.D. Nguyen Thi Khanh Tien

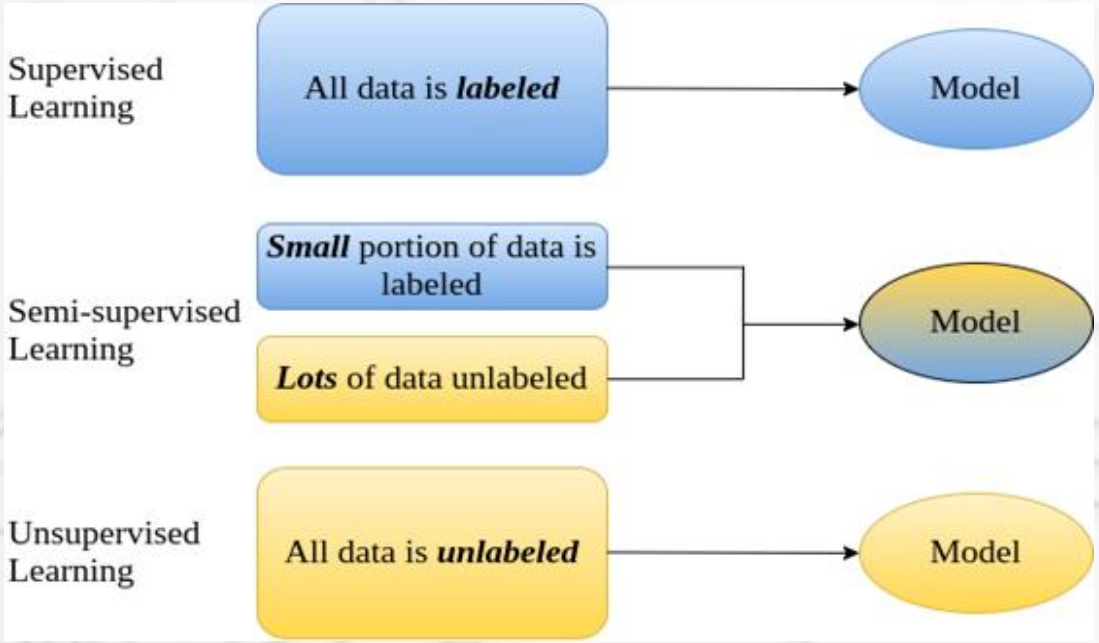
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Introduction to Semi-Supervised Learning

Semi-supervised learning (SSL) is a machine learning technique that leverages both labeled and unlabeled data to train a model. This approach is particularly useful when obtaining large amounts of labeled data is costly or time-consuming. By utilizing unlabeled data, SSL models can often achieve better performance than traditional supervised learning methods.

Key Concepts

- **Labeled Data:** Data points with associated target labels.
- **Unlabeled Data:** Data points without associated target labels.
- **SSL Model:** A machine learning model trained on both labeled and unlabeled data.



How Semi-Supervised Learning Works

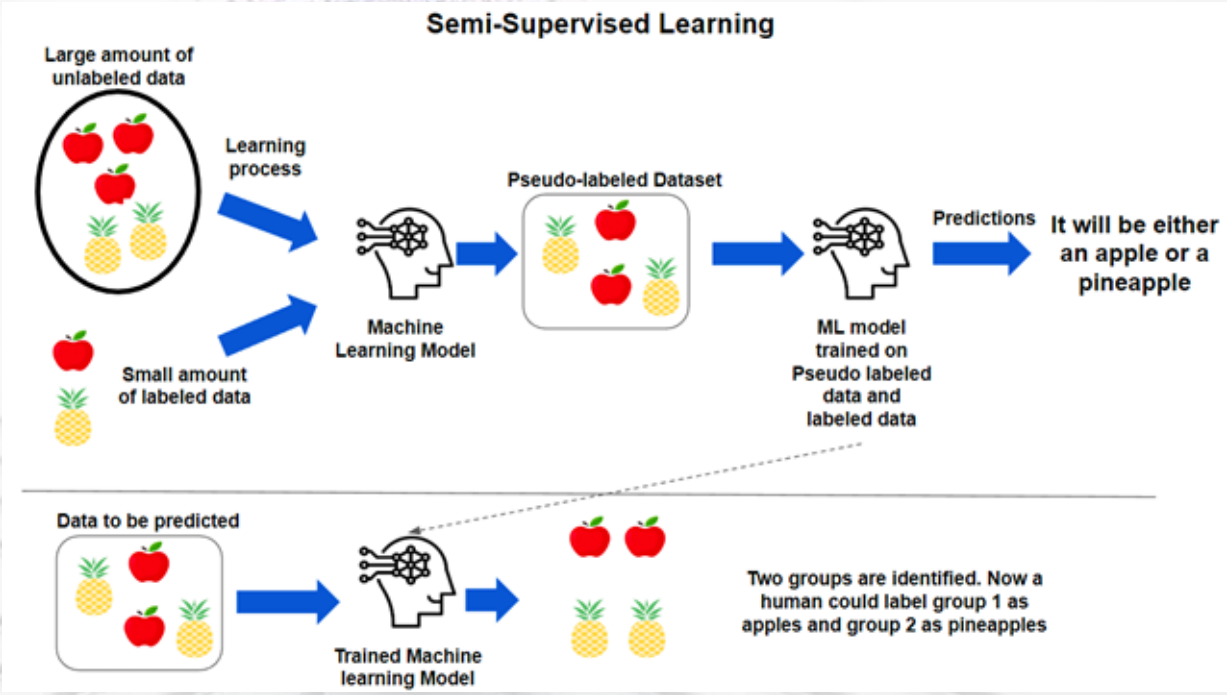
1) Initial Training. The SSL model is initially trained on a small amount of labeled data.

2) Leveraging Unlabeled Data. The model then uses the unlabeled data to learn underlying data distributions and patterns.

3) Iterative Refinement. The model iteratively refines its predictions on the unlabeled data, gradually improving its performance.

4) Label Propagation. The model assigns pseudo-labels to the unlabeled data based on its confidence in the predictions.

5) Model Retraining. The model is retrained on the combined dataset of labeled and pseudo-labeled data.



Introduction to Semi-Supervised Learning

Common Techniques in Semi-Supervised Learning:

1. Self-Training:

- The model makes predictions on unlabeled data.
- The most confident predictions are added to the training set as labeled data.
- The model is retrained on the expanded dataset.

2. Co-Training:

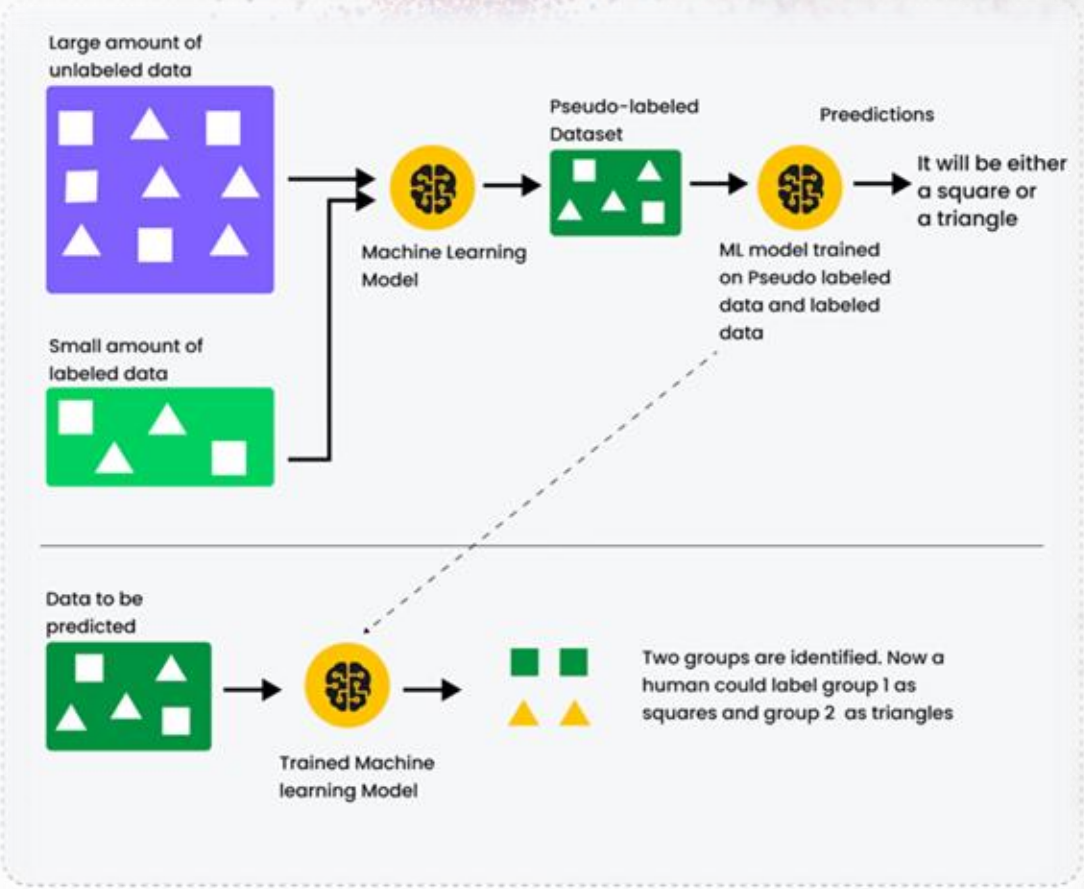
- Multiple models are trained on different views of the data.
- Each model labels unlabeled data, and the labels are used to train the other models.

3. Semi-Supervised Support Vector Machines (SS-SVMs):

- SS-SVMs extend traditional SVMs to incorporate unlabeled data by maximizing the margin between classes while considering the distribution of unlabeled data.

4. Generative Models:

- Generative models, such as Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs), can be used to generate synthetic labeled data.



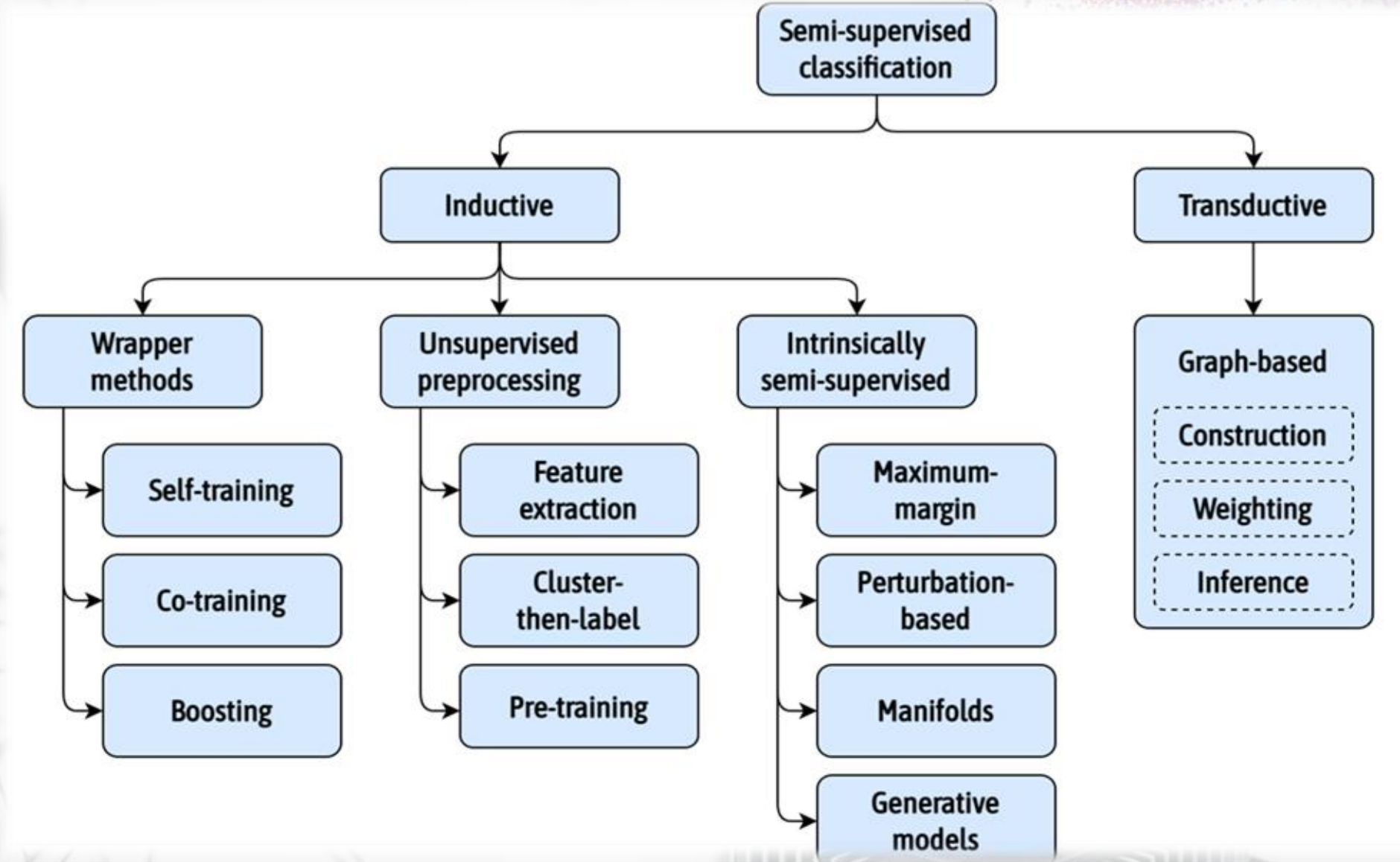
Applications of Semi-Supervised Learning

- **Image Classification:** Training image classifiers with limited labeled data.
- **Natural Language Processing:** Improving language models with large amounts of unlabeled text data.
- **Medical Image Analysis:** Analyzing medical images with limited annotated data.
- **Anomaly Detection:** Identifying outliers in large datasets.

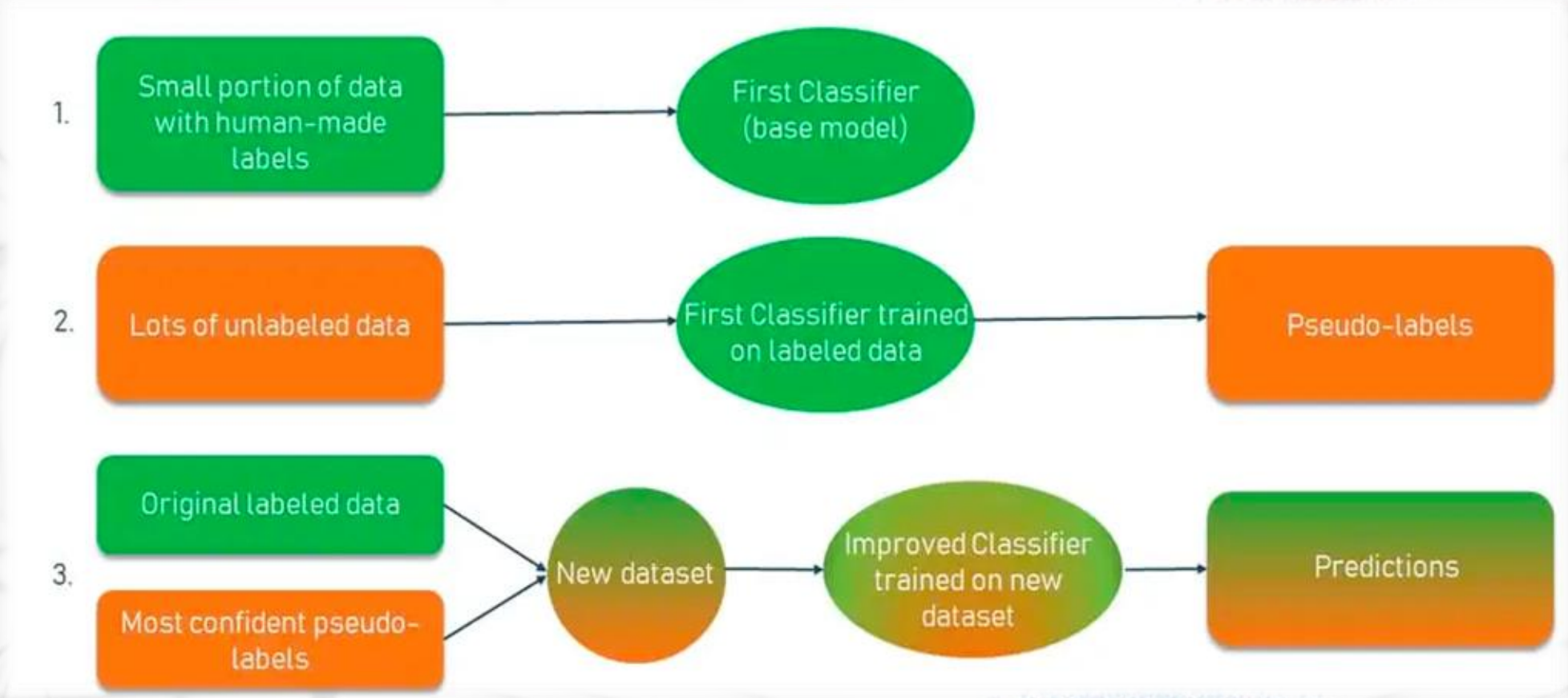
Semi-supervised Learning Potential in Sentiment Analysis



The taxonomy of semi-supervised learning methods

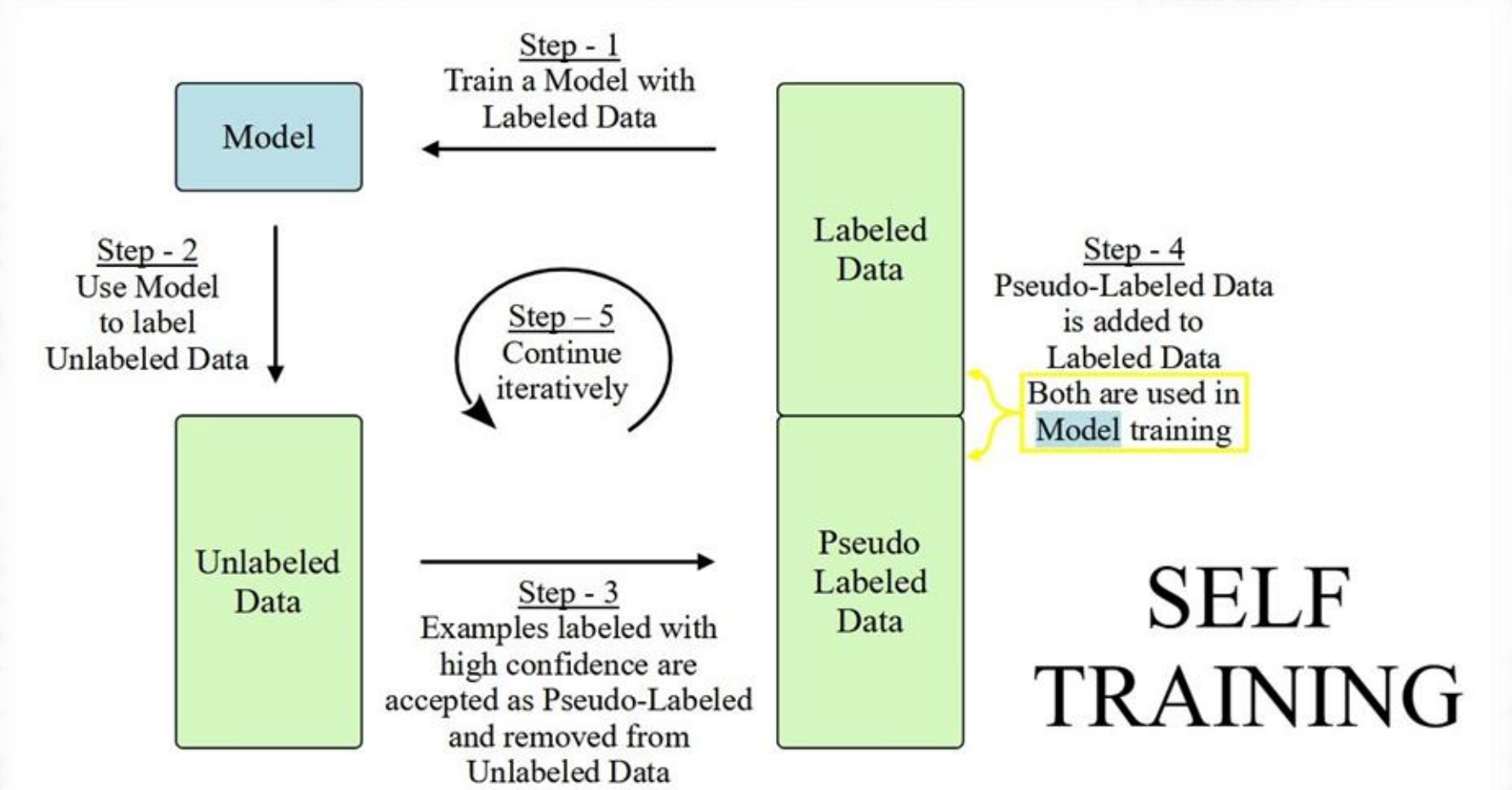


Semi-Supervised Self-Training Method



Self-Training: A Survey → https://arxiv.org/pdf/2202.12040

Semi-Supervised Self-Training Method

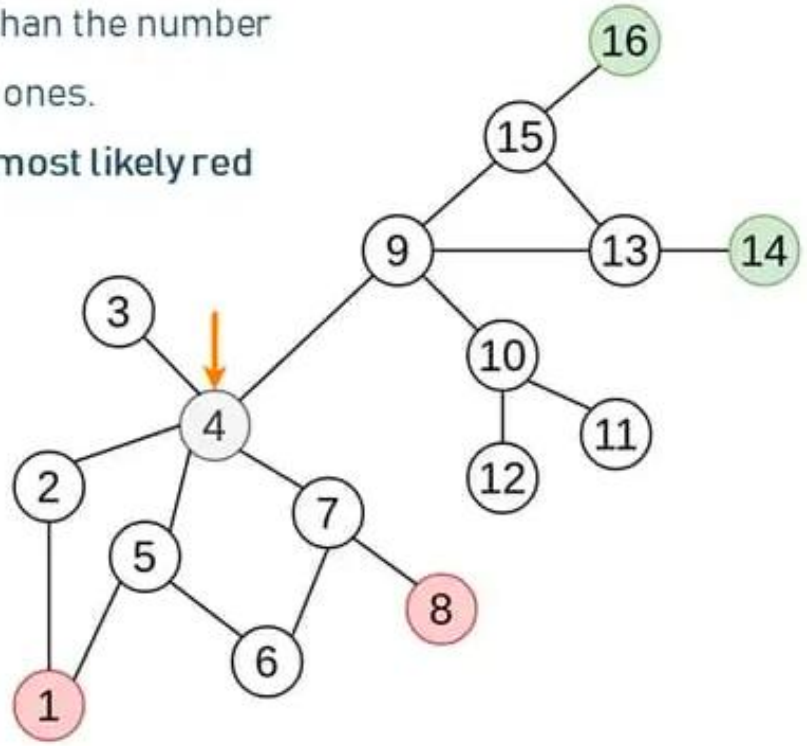


Self-Training: A Survey → https://arxiv.org/pdf/2202.12040

SSL with graph-based label propagation

The number of walks to red nodes is greater than the number of walks to green ones.

Assumption: **4 is most likely red**



Walks that end up in green nodes

1. 4 → 9 → 15 → 16
2. 4 → 9 → 13 → 14
3. 4 → 9 → 13 → 15 → 16
4. 4 → 9 → 15 → 13 → 14

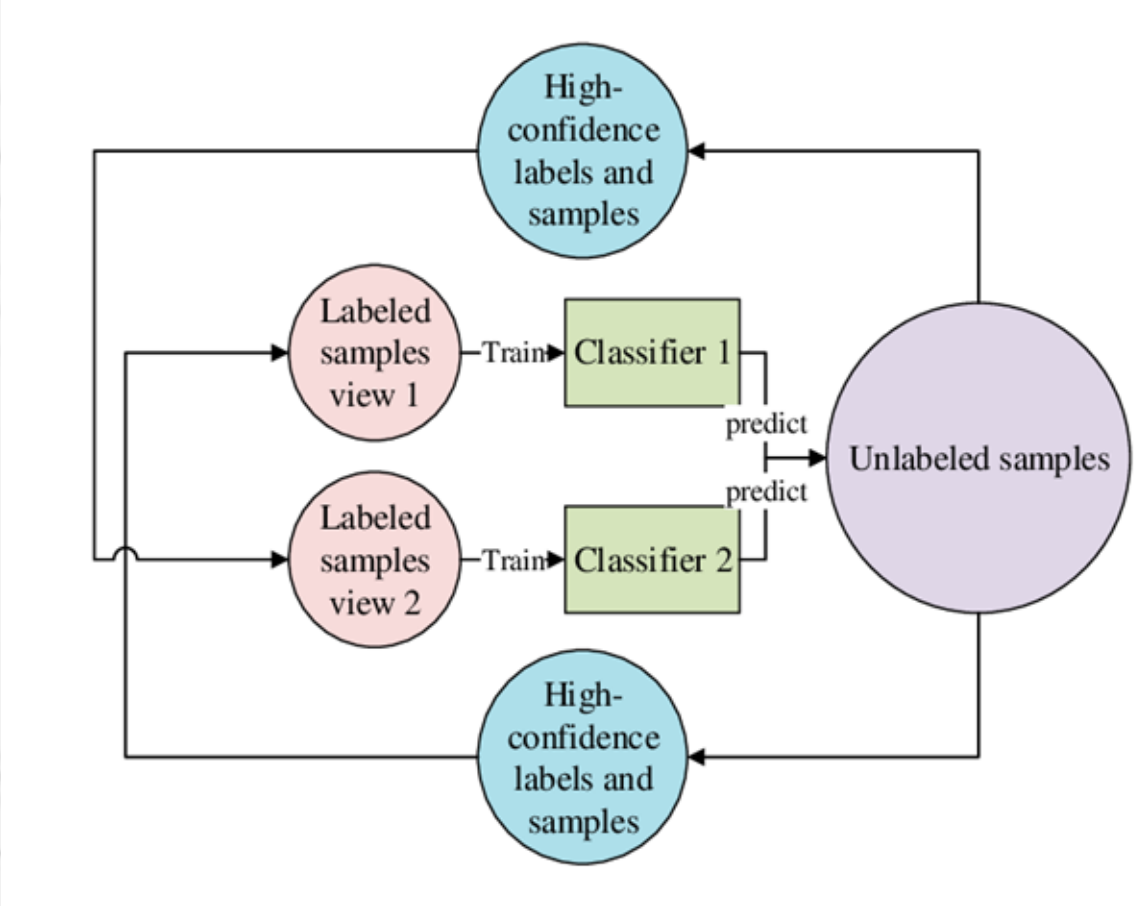
Walks that end up in red nodes

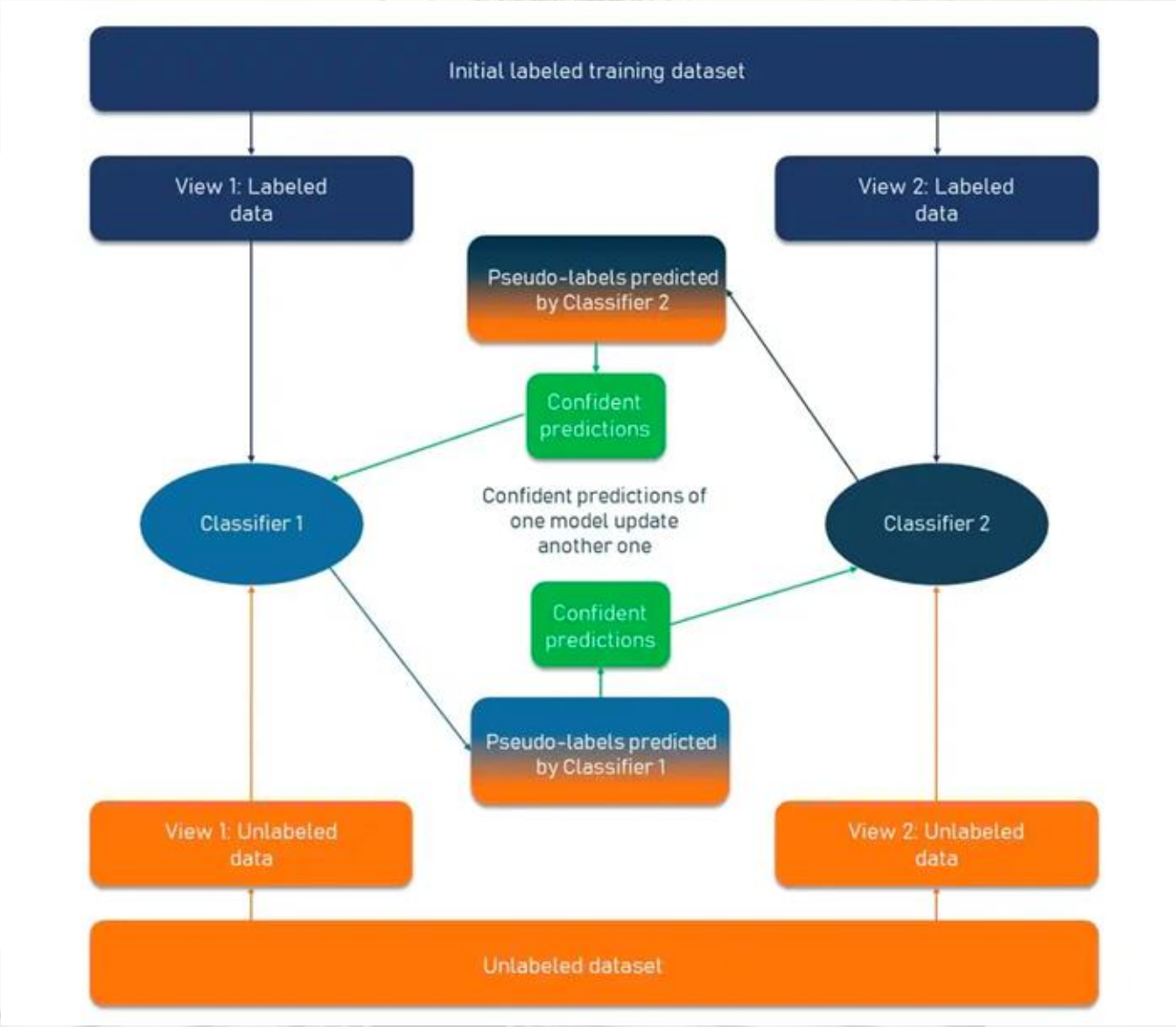
1. 4 → 7 → 8
2. 4 → 7 → 6 → 5 → 1
3. 4 → 5 → 1
4. 4 → 5 → 6 → 7 → 8
5. 4 → 2 → 1

Learning from labeled and unlabeled data with label propagation
→https://pages.cs.wisc.edu/~jerryzhu/pub/CMU-CALD-02-107.pdf

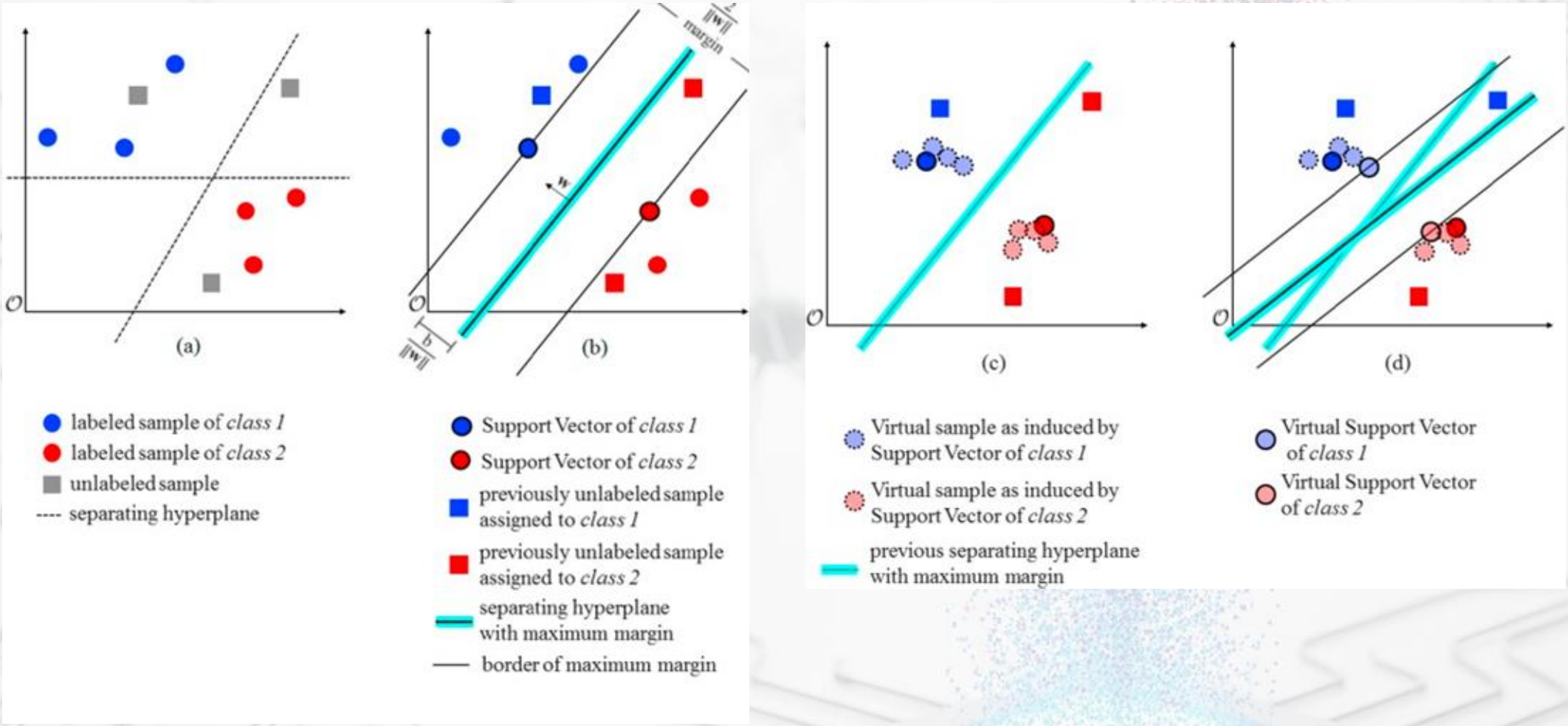
Semi-Supervised Co-Training Method

Co-training research paper
→ https://www.cs.cmu.edu/~avrim/Papers/cotrain.pdf





Semi-Supervised Support Vector Machines



A simple semi-supervised strategy SVM Wine

https://www.kaggle.com/code/saptarsi/a-simple-semi-supervised-strategy-svm-wine?scriptVersionId=46704110